

## Penalty view

$$C_k \rightarrow \lambda_k \rightarrow \Sigma$$

$$\lambda_k = \beta_k \quad (c = 0)$$

exact settings

$\beta_k = \text{const}$

Fixed penalty

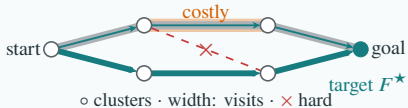
adaptive  $\beta_k$

Lagrangian

## SafeOT-Dual: one price rule

1 Budgeted flow

observed  $\hat{F}$



2 price

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

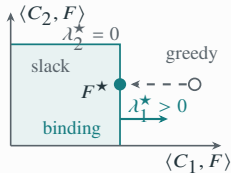
clipped  $c$ , filtered

feedback  $\bar{J}_k - B_k$

3

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A c_k}{1 + \sum_k \lambda_k} \rightarrow \text{PPO/TRPO}$$

## Constraint view



$\varepsilon \rightarrow 0$ : LP dual

clip =  $c$ : active set

conditional limits

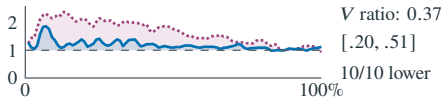
Training cost/ $B$

— SafeOT-Dual

..... TRPO-Lag ( $c = 0$ )

shaded: median excess

SafePO (10 tasks)



Bullet (6 tasks)

